

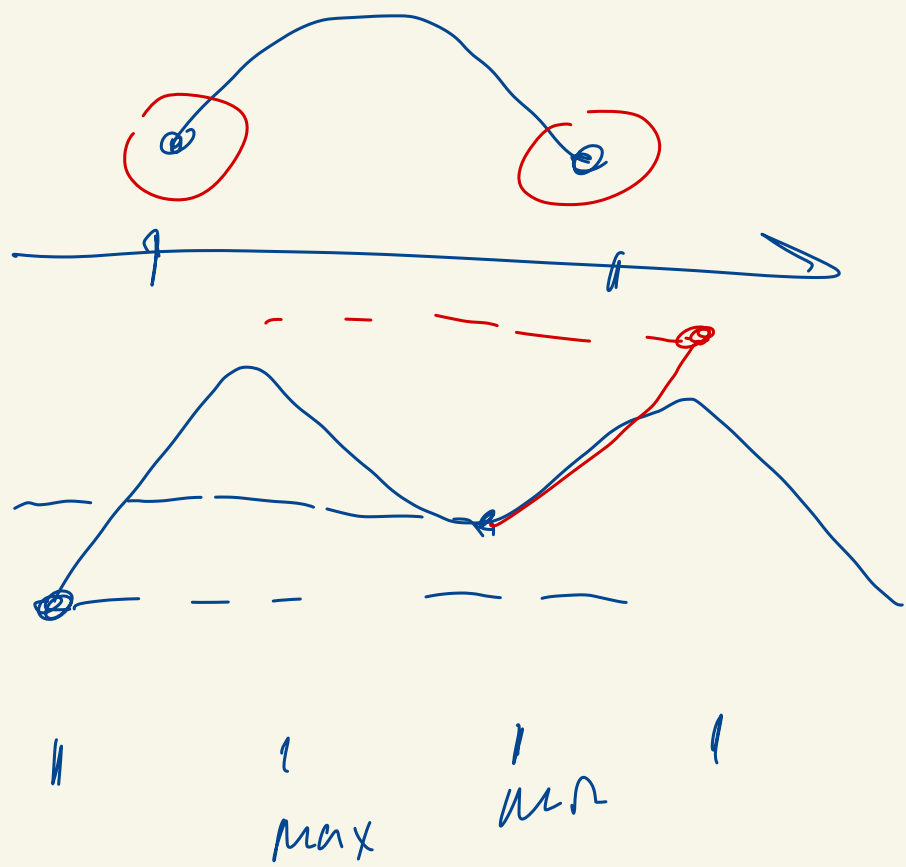


12

$$y(0) = 11$$

$$y(3) = 27 - \underline{18 \cdot 3} + 11$$
$$\underline{\underline{+18}}$$

?



$$\frac{6-x}{4+3x-x^2} > 0$$

Order $[2; 3]$

$$y = a(x-3) + 3$$

$$\begin{cases} 2x^2 + 2y^2 = |x| + |y| \\ \frac{y-3}{x-3} = a \end{cases}$$

$$\begin{cases} x \geq 0 \\ y \geq 0 \end{cases} \rightarrow$$

$$2x^2 - x + 2y^2 - y$$

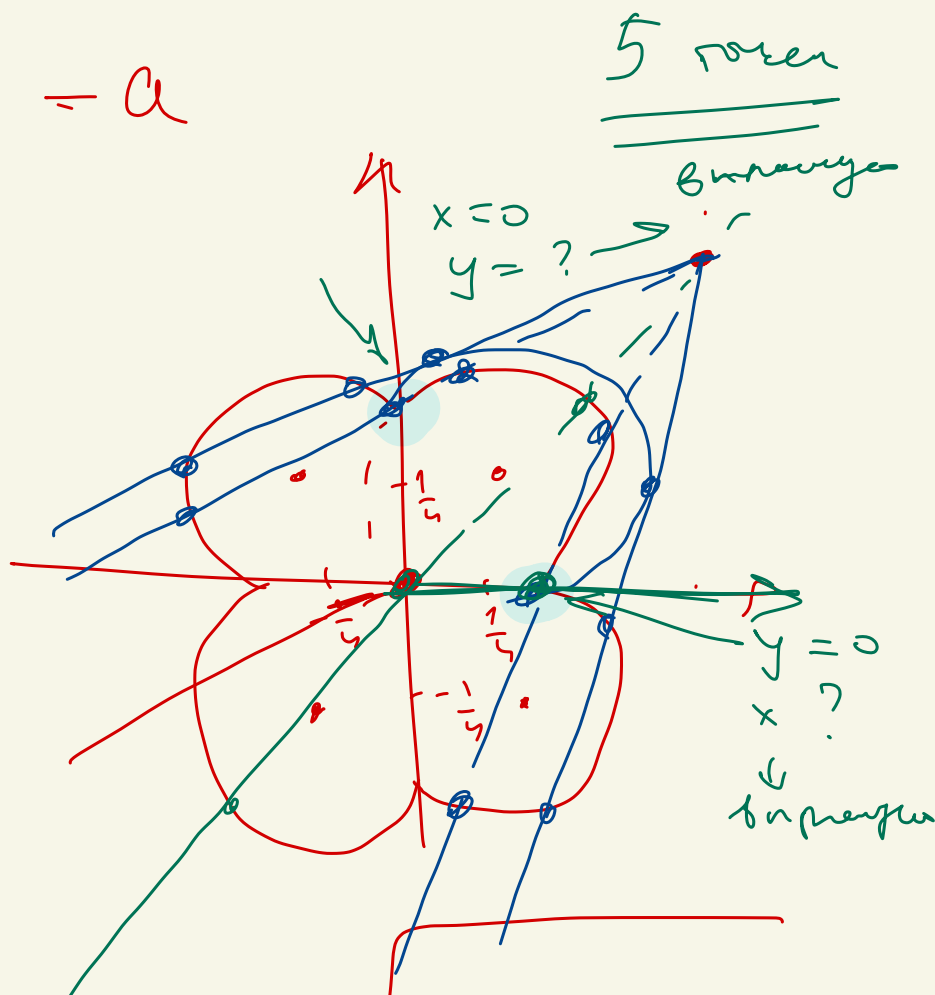
$$2 \left(x^2 - \frac{x}{2} + \right)$$

$$y = a(a-3) + 3$$

$$x \neq 3$$

$$x = 3$$

$$(3, 3)$$



$$2 \left(-\frac{1}{16} \right)$$

$$-\frac{1}{8} - \frac{1}{8} =$$

$$-\frac{2}{8} = -\frac{1}{4}$$

$$\Rightarrow \frac{1}{8} = R^2$$

$$R = \sqrt{\frac{1}{8}}$$

$$2x^2 + 2y^2 = x + y$$

$$x = 0$$

$$2y^2 - y = 0$$

$$y(2y - 1) = 0$$

$$y = 0 \quad y = \frac{1}{2}$$

$$\left(x - \frac{1}{4}\right)^2 + \left(y - \frac{1}{4}\right)^2 = \frac{1}{8}$$

$$\frac{1}{16} + y^2 - \frac{y}{2} + \frac{1}{16} = \frac{1}{8}$$

$$y^2 - \frac{y}{2} = 0$$

$$\frac{2y^2 - y}{2} = 0$$

$$y(2y - 1) = 0$$

$$y = 0 \quad y = \frac{1}{2}$$

$$2y^2 - y = 0$$

$$y = (x - 3)a + 3$$

$$y = \frac{1}{2} \quad x = 0$$

$$a = \frac{5}{6}$$

$$x = \frac{1}{2} \quad y = 0 \quad a = \frac{6}{5}$$

$$2x^2 + 2y^2 = x + y$$

$$y = a(x - 3) + 3$$

$$2x^2 + 2(a^2(x-3)^2 + 9 + 2(a(x-3) + 3)) =$$

$$x + a(x-3) + 3$$

$$2x^2 + \underline{2a^2(x-3)^2} + 18 + 4a(x-3) + 12$$

$$-x - a(x-3) - 3 = 0$$

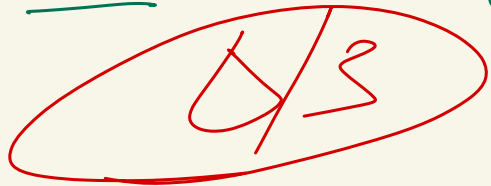
$$2x^2 + \underline{2a^2(x^2 - 6x + 9)} + 27 + \underline{4ax} - \underline{12a}$$

$$\underline{-x} - \underline{ax} + \underline{3a} = 0$$

$$x^2(2 + 2a^2) + x(-12a^2 + 3a - 1) + 18a^2$$

$$-8a + 27 = 0$$

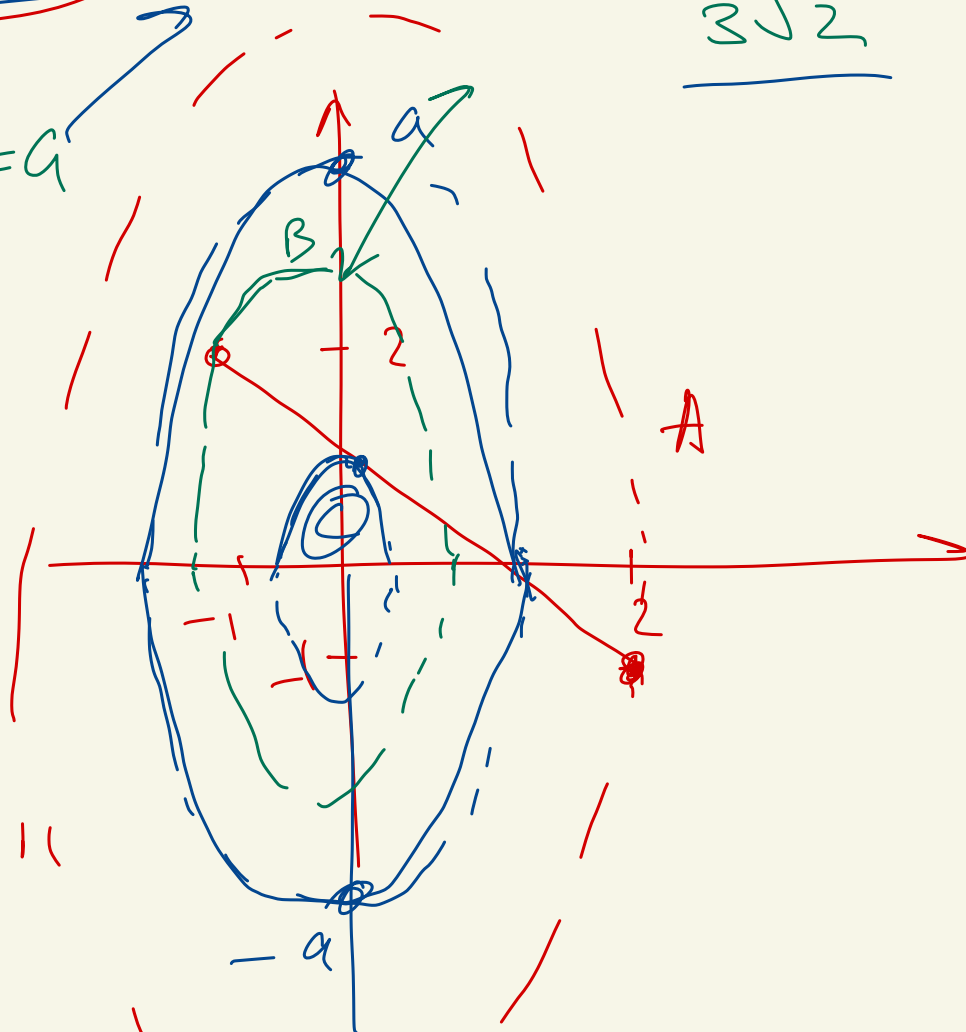
$$D = \underline{(-12a^2 + 3a - 1)^2} - 4 \cdot \underline{2(1+a^2)} \cdot \underline{(18a^2 - 8a + 27)}$$



Уравнение отрезка

$$\sqrt{(x+1)^2 + (y-2)^2} + \sqrt{(x-2)^2 + (y+1)^2} = \underline{\underline{3\sqrt{2}}}$$

$$|y| + x^2 = a$$



$$|y| + x^2 = a$$

$$\underline{\underline{y \geq 0}}$$

$$y = -x^2 + a$$

$$y < 0$$

$$-y + x^2 = a$$

$$y = x^2 - a$$

$$A(2, -1) \Rightarrow$$

$$x^2 - a = y$$

$$4 - a = -1$$

$$\underline{\underline{a = 5}}$$

$$B(-1; 2)$$

$$y = -x^2 + a$$

$$2 = -1 + a$$

$$a = 3$$

$$\underline{\underline{a = 3}}$$

Хитрость:

Знаем, что парабола проходит через отрезок прямой:
 $(-1; 2)$ и $(2; -1)$

$$y = kx + b$$

$$\begin{cases} 2 = -k + b \\ -1 = 2k + b \end{cases}$$

$\Rightarrow$

$$k = b - 2$$

$$-1 = 2(b - 2) + b$$

$$-1 = 2b - 4 + b$$

$$3 = 3b$$

$$b = 1$$

$$k = -1$$

$$y = -x + 1$$

проходит через от

$$-x + 1 = -x^2 + a$$

$$-x^2 + x + a - 1 = 0$$

$$D = 1 + 4(a - 1) = 0$$

$$4a - 4 + 1 = 0$$

$$4a = 3$$

$$a = \frac{3}{4}$$

$$a \in (3; 5] \cup \left\{ \frac{3}{4} \right\}$$

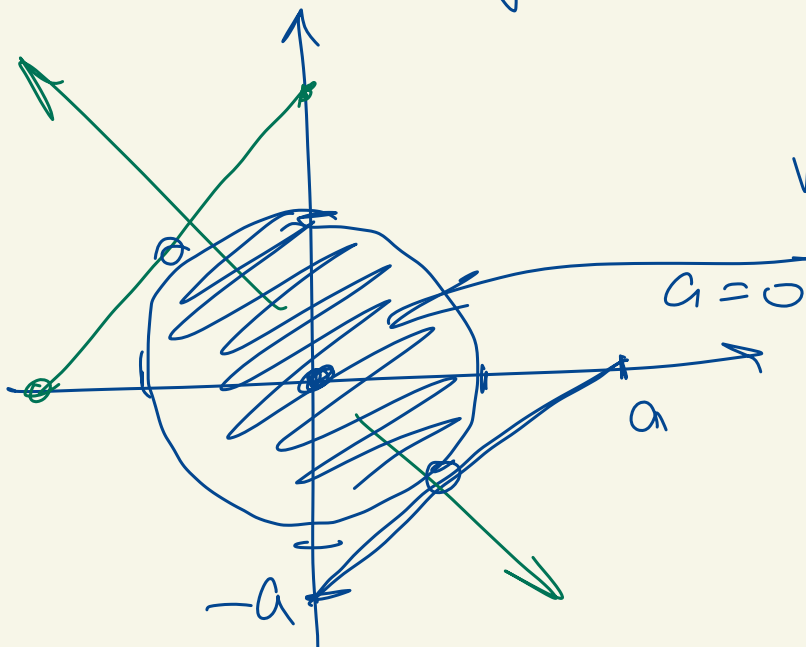
$$\begin{cases} \sqrt{(x-a)^2 + y^2} + \sqrt{x^2 + (y+a)^2} = \underline{\underline{|a\sqrt{2}|}} \\ x^2 + y^2 \leq 18 \end{cases}$$

1 per

$$(a; 0) \quad (0; -a)$$

$$\sqrt{(a-0)^2 + (0-(-a))^2} = \sqrt{a^2 + a^2} =$$

$$\sqrt{2a^2} = \underline{\underline{|a\sqrt{2}|}}$$



$$\sqrt{x^2 + y^2} + \sqrt{x^2 + y^2} = 0$$

$$2\sqrt{x^2 + y^2} = 0$$

$$a=0$$

$$(a \mid 0)$$

$$(0 \mid -a)$$

$$y = kx + b$$

$$\begin{cases} 0 = ak + b \\ -a = 0 + b \end{cases}$$

$$b = -a$$

$$0 = ak - a$$

$$ak = a$$

$$k = 1$$

$$y = x - a$$

$$x^2 + (x-a)^2 = 18$$

$$x^2 + x^2 - 2ax + a^2 = 18$$

$$2x^2 - 2ax + a^2 - 18 = 0$$

$$(2a)^2 - 4 \cdot 2 - (a^2 - 18) = 0$$

$$4a^2 - 8a^2 + 4 \cdot 2 \cdot 18$$

$$-4a^2 + 4 \cdot 2 \cdot 18 = 0$$

$$4h^2 = 4 \cdot 2 \cdot 18$$

$$a^2 = 2 \cdot 18$$

$$a = \pm 6$$

Orbit $a=0$, $a=6$; $a=-6$